Tide retrospective: κ_3 asymptotic for the anharmonic Gibbs

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Abstract

This retrospective records the first *cross-repo* Tide-loop excursion: a single theorem stitching together the abstract κ_3 framework from `timaeus-research/threepoint` and the concrete anharmonic asymptotic machinery from `timaeus-research/laplace`, and proving that for the 1D anharmonic potential $L(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$ with $\lambda, \gamma > 0$ and the discriminant condition $\alpha^2 < 3\lambda\gamma$, the third cumulant of the linear observable in all three slots admits the leading asymptotic $t^2\kappa_3(x, x, x) \to -\alpha/\lambda^3$. The formalisation took 325 lines of Lean in a new file `Laplace/OneD/AnharmonicKappa3.lean`, with zero `sorry` / `axiom` / `native_decide`, and landed in a single session. The decisive contributions of the deliberation step were a numerical sanity check that caught a miscomputed Wick coefficient (Claude wrote $\langle u^6\rangle_0 \cdot \alpha/6 = 3\alpha/2$ when it is in fact $5\alpha/2$), and an architectural recommendation from GPT-5.5 Pro to add `threepoint` as a Lake dependency of `laplace`, keeping the dependency direction generic-to-analytic. The headline result is the actual cross-susceptibility identity the threepoint paper was written for: the cumulant κ_3 at leading order is non-zero precisely because the cubic perturbation breaks the Gaussian parity that makes the harmonic case trivially zero.

1 Setting

The threepoint repository, after Tide 5 (kappa3-harmonic), instantiated its abstract κ_3 framework on the harmonic Gibbs measure and proved the trivial vanishing $\kappa_3(x, x, x) = 0$ by parity. That tide deliberately avoided the `GibbsRegularity` hypothesis bundle[1] because κ_3 is defined at $h = 0$, and at the harmonic level every term in the cumulant expansion contains an odd-power Gaussian moment.

The Tide 5 retrospective flagged the natural next step as "κ_3 for the anharmonic": the cubic term $(\alpha/6)x^3$ in the anharmonic potential breaks parity, so κ_3 is genuinely non-zero, and the formalisation has to do an actual asymptotic calculation rather than read off Wick contractions. The candidates survey of 6 May 2026 listed this as **C2/F1**, a "substantively cross-repo bridge" — the formalisation lives at the boundary between the two seabeds.

The laplace repository, by 6 May, had a long history of asymptotic expansions for the anharmonic Gibbs measure, headlined by `cov_anharmonic_asymptotic`[2]: the first-order asymptotic $\mathrm{Cov}_t[x^2, x] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$ for the same anharmonic potential under the same discriminant condition. Underneath sat the rescaling identity, the J-function asymptotics, and a constellation

[1] It bundles partition positivity, differentiability of the unperturbed partition at $h = 0$, and the integral-of-derivative identities. Required for the FDT and cross-susceptibility *derivative* theorems but not for value-at-zero identities.

[2] From `Laplace/OneD/IntegralRemainder.lean`; the original 27–29 April formalisation.

of polynomial-times-Gaussian moment lemmas. The candidates survey noted that the moment expansions $\langle x\rangle_t, \langle x^2\rangle_t, \langle x^3\rangle_t$ ought to have leading-order forms that combine cleanly with $\mathrm{Cov}_t[x^2, x]$ to give the κ_3 asymptotic.

2 The thing formalised

The headline statement is:

> *For $\lambda > 0$, $\gamma > 0$, $\alpha^2 < 3\lambda\gamma$, the 1D anharmonic potential $L(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$ on $\mathbb{R}$ against Lebesgue measure, and the linear observable $\varphi(x) = A(x) = B(x) = x$ in all three slots,*
>
> $$t^2 \cdot \kappa_3\big(\mathrm{vol}, L, x, t, x, x\big) \longrightarrow -\frac{\alpha}{\lambda^3} \qquad \textit{as } t \to \infty.$$

The Lean signature (in `Laplace/OneD/AnharmonicKappa3.lean`[3]):

```
theorem kappa3_anharmonic_id_id_id_asymptotic
    {lam alpha gamma : ℝ}
    (hlam : 0 < lam) (hgamma : 0 < gamma)
    (hdisc : alpha ^ 2 < 3 * lam * gamma) :
    Filter.Tendsto
      (fun t : ℝ => t ^ 2 * Threepoint.kappa3 (volume : Measure ℝ)
          (anharmonicPotential lam alpha gamma)
          (fun x : ℝ => x) t (fun x : ℝ => x) (fun x : ℝ => x))
      Filter.atTop
      (nhds (-alpha / lam ^ 3))
```

The discriminant condition $\alpha^2 < 3\lambda\gamma$ is the same one that governs the existing covariance asymptotic.[4] It ensures L is globally coercive, so the Gibbs measure is well-defined and the Laplace asymptotic is dominated by $x = 0$. The hypothesis was the load-bearing correction from a much earlier GPT consult on the original anharmonic formalisation; here it is inherited unchanged.

The instantiation $\varphi = A = B = x$ in all three slots of κ_3 matches the harmonic statement from Tide 5:[5] the "id, id, id" naming refers to all three observables, including the perturbation observable A, being the identity. (A subtle conceptual point caught mid-formalisation: the A parameter does not drop out of the cumulant expansion at $h = 0$, because A also appears as a normal observable inside terms like $\langle A \cdot B\rangle \cdot \langle\varphi\rangle$. See §**??**.)

3 Proof strategy

The proof factors through a polynomial identity in the moments and three asymptotic theorems already proven in laplace. The chain of reasoning is short enough to write out:

[3] Permalink: `Laplace/OneD/AnharmonicKappa3.lean` on `tide/anharmonic-kappa3` at commit `8a62893`.

[4] `cov_anharmonic_asymptotic` in `IntegralRemainder.lean`.

[5] `Threepoint.kappa3_harmonic_id_id_id_eq_zero`.

The algebraic identity. For any measure and any potential,

$$\kappa_3(x,x,x) \;=\; \mathrm{Cov}[x^2,x] - 2\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3.$$

Verification by expansion: $\mathrm{Cov}[x^2,x] = \langle x^3\rangle - \langle x^2\rangle\langle x\rangle$, so the right-hand side is $\langle x^3\rangle - \langle x^2\rangle\langle x\rangle - 2\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3 = \langle x^3\rangle - 3\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3 = \kappa_3(x,x,x)$. This is a pure ring identity in the three moments; in Lean it is one `ring` call after expanding `gibbsCov`.

The three asymptotics. All three exist as user-facing theorems in laplace's integral-remainder module.[6]

- `cov_anharmonic_asymptotic`: $t^2 \cdot \mathrm{Cov}_t[x^2,x] \to -2\alpha/\lambda^3$.
- `mean_anharmonic_asymptotic`: $t \cdot \langle x\rangle_t \to -\alpha/(2\lambda^2)$.
- `cov_self_anharmonic_asymptotic`: $t \cdot \mathrm{Var}[x]_t \to 1/\lambda$ (stated as $\mathrm{Cov}_t[x,x]$, i.e. the variance).

One small new asymptotic. The bridge needs $t \cdot \langle x^2\rangle_t \to 1/\lambda$, which is not directly stated in laplace but follows from the existing variance and mean asymptotics via $\langle x^2\rangle = \mathrm{Var}[x] + \langle x\rangle^2$. The mean-square term $\langle x\rangle^2$ is $O(1/t^2)$ (since $t \cdot \langle x\rangle$ is bounded), so $t \cdot \langle x\rangle^2$ is $O(1/t) \to 0$; combine with $t \cdot \mathrm{Var}[x] \to 1/\lambda$ via `Tendsto.add`. This is packaged as `secondMoment_anharmonic_asymptotic`, 25 lines, exposed at the top level for downstream reuse.

Combining. The headline theorem unfolds the seven `Threepoint.gibbsExp ... 0` terms inside κ_3 to `Laplace.gibbsExpectation` (one bridging lemma; see §??), applies the algebraic identity, and combines the three asymptotics via `Tendsto.add`, `Tendsto.mul`, and `Tendsto.const_mul`. The terminal limit-value computation is one `field_simp + ring`:

$$-\tfrac{2\alpha}{\lambda^3} - 2 \cdot \tfrac{1}{\lambda} \cdot \big(-\tfrac{\alpha}{2\lambda^2}\big) + 0 \;=\; -\tfrac{2\alpha}{\lambda^3} + \tfrac{\alpha}{\lambda^3} \;=\; -\tfrac{\alpha}{\lambda^3}.$$

The mathematical content is roughly 30 lines; the rest of the 325-line file is the four bridging lemmas and the `Tendsto`-algebra plumbing.

4 Roadblocks and resolutions

The miscomputed Wick coefficient. Pre-deliberation Claude calculated

$$\langle x^3\rangle_t \approx -\tfrac{3\alpha}{2\lambda^3 t^2},$$

which gives a κ_3 leading order of $0 \cdot t^{-2} + O(t^{-3})$ — a vanishing claim. The actual coefficient is $5\alpha/(2\lambda^3 t^2)$, because $\langle u^6\rangle_0 = 5!! = 15$ in the rescaled Gaussian, giving the prefactor $15/6 = 5/2$ rather than $3/2$. The leading-order κ_3 is then $-\alpha/\lambda^3 \cdot t^{-2}$, *not* zero.

The error was caught *before* the GPT consult by running a small `scipy.integrate.quad` sanity check[7] at $\lambda = 1, \alpha = 0.5, \gamma = 1$ and $t \in \{4, 8, \ldots, 256\}$. The numerical $\kappa_3 \cdot t^2$ converges to -0.495, matching the corrected prediction $-\alpha/\lambda^3 = -0.5$ to four significant figures. GPT-5.5 Pro independently arrived at the same correction with the same 5/2 vs. 3/2 diagnosis, which gave a high-confidence double-check before any Lean was attempted.

[6] Path: `Laplace/OneD/IntegralRemainder.lean`. See `retrospective.tex` (§6–7) for the originals.

[7] Saved at `sri/projects/patterning/staging/anharmonic-kappa3-numerics.py`.

> **GPT-5.5 Pro:** Candidate A is not correct. The leading t^{-2} term of $\kappa_3(x,x,x)$ does not vanish. The correct leading asymptotic is $\kappa_3(x,x,x) \sim -\alpha/\lambda^3 \cdot t^{-2}$. ... You wrote $\langle x^3\rangle_t = -\frac{3\alpha}{2\lambda^3}t^{-2} + o(t^{-2})$, but the correct leading term is $\langle x^3\rangle_t = -\frac{5\alpha}{2\lambda^3}t^{-2} + o(t^{-2})$.

The lesson generalises: *always run the numerical sanity check before the GPT consult, even when the Lean machinery is mostly already in place.* The skill's discipline anticipates this; the Wick coefficient trap is a concrete instance.

The architectural question. The candidates survey listed three options for where the new theorem should live: in threepoint (re-deriving the anharmonic asymptotics inline), in laplace (with a local copy of κ_3 to avoid the cross-repo dependency), or with laplace adding a `[[require]]` on threepoint. GPT-5.5 Pro recommended the third unambiguously:

> **GPT-5.5 Pro: Put the theorem in `laplace`, and make `laplace` depend on `threepoint` if feasible.** Reason: `threepoint` owns the generic cumulant notion. `laplace` owns the anharmonic asymptotic machinery. So the natural layering is generic → analytic application, i.e. `laplace` imports `threepoint`, not the other way around.

The dependency was added by editing the lakefile[8] of laplace to point at threepoint at a fixed commit. The threepoint repository builds in ∼30 s on first import, ∼8 s incrementally, and shares laplace's Mathlib v4.29.0 pin. Setup took about three minutes of wall-clock; the build took the rest.

The A-parameter snag. An early version of the unfolding lemma left the perturbation observable A abstract[9] and tried to prove that this equals $\langle x^3\rangle - 3\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3$. This is wrong: the A parameter is not just the perturbation observable that drops out at $h = 0$; it also appears as a normal observable inside terms like $\langle AB\rangle\langle\varphi\rangle$ in the κ_3 expansion. The fix was to set $A = (\text{fun } x \to x)$ explicitly — which is what the harmonic instantiation in threepoint[10] does too; the "id, id, id" naming refers to all three slots, perturbation observable included. With $A = (\text{fun } x \to x)$, the A contributions in the expansion become further $\langle x\rangle$ factors and the polynomial identity goes through.

Rewriting under the integrand lambda. The unfolding lemma converts seven perturbed-Gibbs-expectation terms[11] to laplace's `gibbsExpectation`. After unfolding the cumulant, the integrand lambdas appear with a beta-redex,[12] not already canonicalised to w form. Two failed approaches: rewriting via the bridge lemma directly[13]; and rewriting the integrand shape via a function-extensionality lemma[14]. The fix: keep the bridge lemma's RHS in `gibbsExpectation` form (don't unfold past it), and match the integrand-lambda shape including the beta-redex literally:

[8] Add a `[[require]]` block with `name = "Threepoint"`, `git = "..."`, `rev = "40398d8"`, then run `lake update Threepoint`.

[9] Concretely, `Threepoint.kappa3 volume L A t (fun x => x) (fun x => x)` with generic A.

[10] `Threepoint.kappa3_harmonic_id_id_id_eq_zero`.

[11] `Threepoint.gibbsExp ... 0` for each of the seven κ_3 summands.

[12] Concretely: `(fun x => x) w * A w * (fun x => x) w` and similar.

[13] `rw` with the bridging lemma `threepoint_gibbsExp_volume_zero_eq`; failed because the goal had already been auto-converted into the ratio-of-integrals form.

[14] `rw [h_cube]` with `h_cube : (fun w => w * w * w) = (fun w => w ^ 3)`; failed because the integrand inside the integral was inlined, not in lambda form.

```
have h_cube : (fun w => (fun x => x) w * (fun x => x) w
                * (fun x => x) w) = (fun w => w ^ 3) := by
  funext w; ring
```

This is the kind of friction the laplace `CLAUDE.md` should record; added to the followups.

The `nhds (...)` congruence shuffle. Three places in the proof construct a `Tendsto` via `Tendsto.mul` or `Tendsto.add` and arrive at a limit value like `nhds (1/lam + 0)` or `nhds (2 * C * C * 0)`, which Lean does *not* simplify to `nhds (1/lam)` or `nhds 0` automatically. The fix is a one-line `have h_lim_eq : ...= ...:= by ring` followed by `rw [h_lim_eq] at h_prod`. Three instances; documented.

Mathlib name gotcha for $1/t \to 0$. The expected name does not exist;[15] the bridge is `by simp only [one_div]; exact tendsto_inv_atTop_zero`.

5 What was Mathlib, what was new

Mathlib provided.

- `tendsto_inv_atTop_zero`: $1/t \to 0$ as $t \to \infty$ (after a `one_div` rewrite).
- `Tendsto.mul`, `Tendsto.add`, `Tendsto.const_mul`, `Tendsto.congr'`, `tendsto_const_nhds`: standard `Tendsto`-algebra glue.
- `Filter.eventually_gt_atTop`: "eventually $t > 0$" under `atTop`, used to safely `field_simp` an $1/t \cdot t = 1$ equality.
- `ring`, `field_simp`, `funext`: the usual.

Threepoint provided.

- `Threepoint.kappa3`: the abstract third cumulant, defined generically over a measurable space (W, μ) and a perturbed Gibbs expectation `gibbsExp` $\mu\ L\ A\ t\ h\ \varphi$.
- `Threepoint.gibbsExp`: the perturbed Gibbs expectation, specialised at $h = 0$ to the unperturbed ratio of integrals.

Laplace provided. The three asymptotics named in §**??**, plus the original `anharmonicPotential`, `gibbsExpectation`, and `gibbsCov` definitions.

What was new.

- `threepoint_gibbsExp_volume_zero_eq`: bridges `Threepoint.gibbsExp volume L A t 0` $\varphi =$ `Laplace.gibbsExpectation L t` φ. Three lines: unfold both, simp on the perturbation factor.
- `kappa3_id_id_id_unfold`: unfolds `kappa3 volume L x t x x` to $\langle x^3\rangle - 3\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3$ in laplace-side moments. Twenty lines including the integrand-lambda gymnastics.

[15] Searched-for: `tendsto_one_div_atTop_nhds_zero`. Actual: `tendsto_inv_atTop_zero` (uses x^{-1} rather than $1/x$).

- `kappa3_id_id_id_eq_cov_form`: rewrites the three-moment expansion in covariance form, $\mathrm{Cov}[x^2, x] - 2\langle x^2\rangle\langle x\rangle + 2\langle x\rangle^3$. Pure ring after expanding `gibbsCov`; eight lines.
- `secondMoment_anharmonic_asymptotic`: the $t \cdot \langle x^2\rangle \to 1/\lambda$ corollary of the existing mean and variance asymptotics. Forty lines including the proof that $\langle x\rangle \to 0$.
- The headline theorem itself: ∼80 lines of `Tendsto` algebra.

The mathematical content is the algebraic identity (which is elementary) and the limit-value calculation $-2\alpha/\lambda^3 + \alpha/\lambda^3 = -\alpha/\lambda^3$ (which is `ring`). Everything else is plumbing.

6 Lessons

- **Numerical sanity check first, GPT consult second.** The Wick coefficient error would have wasted multiple hours of formalisation if it had survived to Step 3. The 30-line numpy script took two minutes to write and caught the error decisively. The skill's discipline says to do the numerical check; the lesson here is that "before GPT" is the right time, not after.
- **Cross-repo Tide loops are tractable when the dependency direction is clean.** Adding `threepoint` as a Lake dependency of `laplace` took three minutes; the build was ∼30 s on first import. The architectural question (where does the theorem live?) was the only non-trivial decision, and GPT-5.5 Pro resolved it cleanly by appealing to the generic-to-analytic dependency direction.
- **The A parameter in `Threepoint.kappa3` does not drop out at $h = 0$.** It also appears as a normal observable inside the cumulant expansion. "id, id, id" includes the A slot, paralleling `Threepoint.kappa3_harmonic_id_id_id_eq_zero`. Worth landing in `Threepoint/CLAUDE.md` as a recurring trap.
- **The integrand-lambda gymnastics are a recurring laplace friction.** Three of the four new lemmas in this file ended up doing `funext w; ring` rewrites of `(fun x => x) w * ...` to $w \cdot \ldots$ to canonicalise integrand shapes. Add to `laplace/CLAUDE.md`: *when an unfolded `gibbsExpectation` has `(fun x => x) w` beta-redexes in its integrand, match them literally in the rewrite hypotheses; do not expect `ring` to see through them.*
- **`nhds (a + 0)` is not `nhds a` syntactically.** The clean idiom is to pre-compute the simplified limit-value as a `have h_lim : a + 0 = a := by ring` and `rw [h_lim] at hSum` before the `congr'`. Recurs three times in this file. Worth a generic helper or a tactic note in `laplace/CLAUDE.md`.
- **Numerical sanity scales: a 30-line scipy script doubles as the closed-form expectation in the tide log.** Recording $\kappa_3 \cdot t^2 \approx -0.495$ at $t = 256, \lambda = 1, \alpha = 0.5$ in the tide log gives the closed-form expectation; running the same script at the same parameters during the formalisation gives the sanity check; the agreement to four significant figures is the post-formalisation cross-check. Three uses of one artefact.

7 Follow-ups

- *Strict-improvement: the $\langle x^3\rangle_t$ asymptotic as a corollary.* GPT noted that with the moment expansions exposed, $t^2\langle x^3\rangle_t \to -5\alpha/(2\lambda^3)$ is a ∼5-line corollary of $\mathrm{Cov}[x^2, x] = \langle x^3\rangle - \langle x^2\rangle\langle x\rangle$

plus the existing asymptotics. Worth packaging as `thirdMoment_anharmonic_asymptotic` for downstream consistency and as a user-facing version of an asymptotic that currently exists only inside intermediate proofs.

- *Strict-improvement: the same theorem for affine observables (scalar-multiple in each slot).* Generalise from $(\varphi, A, B) = (x, x, x)$ to $(b\,x, a\,x, c\,x)$. Multilinearity of κ_3 extracts the factor abc, and the same parity / asymptotic arguments apply. About 40 lines on top of the present file. Parallels *Candidate* A' from the Tide 5 retrospective.

- *Next-order asymptotic.* The t^{-3} correction to $\kappa_3(x, x, x)$ has an explicit constant $C(\lambda, \alpha, \gamma)$ that requires the moment expansions to one higher order. Substantially larger than the present tide; would benefit from first packaging the mean and second-moment expansions to one more order in `IntegralRemainder.lean`.

- *Promote the algebraic identity to threepoint.* Currently `kappa3_id_id_id_eq_cov_form` is private to laplace. If a second use materialises (e.g. a κ_3 asymptotic on the quartic-with-cubic potential, or a different choice of three identical observables), promote it to a small `Threepoint/Algebra.lean` file. The eight-line proof transfers verbatim.

- *The lean-common promotion candidate.* The "$\langle x^2\rangle = \mathrm{Var}[x] + \langle x\rangle^2$ at the moment-asymptotic level" construction inside the second-moment helper is generic — it works for any potential where the variance and mean asymptotics are given. Worth a generic helper[16] in `lean-common` once a second use materialises.

- *Tag the threepoint paper with `\leanref`.* The threepoint paper currently has no `\leanref` tags pointing at this result. Once the next non-trivial threepoint formalisation lands, do a one-pass tagging consistent with the laplace primer setup.

- *Add the new gotchas to `laplace/CLAUDE.md`.* Three items: the `nhds (a + 0)` normalisation pattern, the integrand-lambda `(fun x => x) w` matching idiom, and the "A does not drop out at $h = 0$" note for cross-repo κ_3 work.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the independent rederivation of the corrected $5\alpha/2$ Wick coefficient, the architectural recommendation to layer `laplace` on top of `threepoint`, and the cleanest algebraic-shortcut form of the proof. The full consult and the tide log itself (with candidate drafts, vote, numerical check, and result) are preserved in the patterning project's tide-log directory.[17]

The seabed for this tide spans the original few-day laplace formalisation (whose anharmonic covariance theorem and multivariate machinery were the load-bearing prior work[18]) and the three-point scaffold from Tide 5 (whose harmonic-κ_3 theorem[19] gave the proof template and the "id, id, id" naming convention). The present tide is small only because so much was already in place.

[16] Tentative name: `secondMoment_eq_var_plus_mean_sq_tendsto`.

[17] `sri/projects/patterning/tide-log/2026-05-06-tide-anharmonic-kappa3.md` and `gpt55_tide_anharmonic_kappa3_v1.md`.

[18] `cov_anharmonic_asymptotic` from `retrospective.tex` and the multivariate sharp track from `retrospective2.tex`.

[19] `kappa3_harmonic_id_id_id_eq_zero`.